ARVENAIRE / ENGINEERING PATHWAYS

Graduate and PhD Career Roadmap

Germany | Japan | South Korea

A practical guide for your next decision
For engineering graduates and researchers comparing a master's, doctorate, funded research position, or later industry role. The country notes are prompts for checking official program rules, not admissions or scholarship promises.

Use the live instructions of the employer, university, scholarship body and immigration authority for your own case.

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Text is selectable and searchable. Headings and reading order are kept simple, with high contrast. This PDF is not tagged for assistive technology and has no fillable fields.

01 / DEGREE, RESEARCH AND CAREER

Choose the route before choosing a country

A research degree is a sequence of academic fit, admission, funding, residence permission and career planning. The sequence differs by institution and country. Separate a degree place from a scholarship, and separate a research position from a non-degree preparatory status.

Country	What to verify	Decision check
Germany	DAAD describes individual and structured doctoral routes. Individual research may take place at a university, research organization or in industry; structured programs have their own application path. Find a supervisor or vacancy and verify the host's rules. [1, 2, 3]	Compare the exact position, supervision, funding term and work or study status.
Japan	Graduate schools differ in whether applicants apply directly or first take a research-student route. That research-student status may be non-degree. MEXT research scholarships have embassy and university recommendation routes; current country and university guidance controls. [5, 6]	Ask whether the offer is a degree place, a non-degree research period or a transition route.
South Korea	Study in Korea describes master's and doctoral study. Check the university and department's live admission criteria and dates. Its published 2026 GKS graduate notice shows embassy and university tracks; use the newest notice and designated university list. [7, 8]	Check the department, degree level, scholarship track, documents and cycle separately.

The official call beats a general summary

A department's current admission page, scholarship guideline and immigration authority decide the actual eligibility, documents, dates and procedure. Do not reuse a previous cycle's calendar.

02 / SHORTLIST LABS AND PROGRAMS

Find a research fit you can explain

Start with the engineering question you want to investigate. Then find groups where the current work, equipment, supervision and funding make that question feasible. Your fit notes should cite a current official lab, university, research-center or position page.

Fit question	Record a concrete answer
Research question	What physical system, process, material, control problem or measurement do I want to understand?
Group fit	Which recent work from this group is actually related? Name a paper, project or facility and your connection.
Method fit	Which method can I already use, and what must I learn? State evidence and gaps honestly.
Supervisor / route	Does the position invite direct applications, require supervisor contact, use a central portal or only accept a scholarship track?
Resources	Are equipment, data, collaborators and access described? What should I ask before accepting?
Funding and terms	What is funded, for how long, under what contract or scholarship, and what costs remain?
Language and output	What language is used for coursework, lab work, supervision, publications and later roles?

Scorecard without a fake score

For each program, mark each item as **verified**, **unclear** or **not a fit**: question fit; host and supervisor; degree status; funding; current admissions path; language; residence route; next official deadline. Do not average these into a numerical ranking. A missing funding term or unclear degree status deserves a direct question.

> **Contact a supervisor only when the route supports it**
> If the program invites supervisor contact, write a short individualized note: why this group, what evidence you bring, what research question you want to discuss, and which formal call you are following. Follow its stated contact channel and documents. No message template can promise a reply or admission.

03 / PREPARE ONCE, TAILOR EACH TIME

Build the application evidence pack

Use one organized master folder, but submit only the current program's requested items. Official Japanese graduate-school guidance lists common documents such as prior degree records, transcripts, recommendations, a thesis/abstract and research proposal while warning applicants to check each school's requirements. This is an example of variation, not a universal checklist. [5]

Pack section	Check
Academic record	Degree certificates or expected-completion proof, transcripts, grading scale and any required credential authentication.
Research record	CV, thesis abstract, research statement/proposal, selected paper/poster/code or design evidence.
References	Letters in the prescribed form, from the permitted referees, sent through the requested channel.
Language and tests	Only the scores and language evidence required by the specific program or scholarship.
Identity / forms	Application form, passport-name consistency, photo or other identity item if the current notice requests it.
Translations	Required language, translation, notarization or legalization rules from the current official call.
Fit statement	Why this exact lab or program; what question and method; what you contribute; what you need to learn.

Work backwards from the official deadline

- Copy the deadline and time zone from the current university or scholarship notice.
- Confirm which documents need a referee, issuing office, translation or authentication.
- Ask each referee early and send the exact instructions and deadline.
- Prepare the research statement for the program's field, page limit and assessment criteria.
- Check that online forms, file names, portal uploads and delivery method are complete.

Separate admission from scholarship selection

A university may admit a student without awarding a scholarship, and a scholarship route can have a separate nomination or screening process. Verify both decisions, conditions and timing with the current official sources.

04 / KNOW WHAT AN OFFER COVERS

Compare funding and conditions

Do not compare offers from stipend headlines alone. Write down what the institution or funder confirms in writing. Germany's DAAD points applicants to employment, scholarships and other funding sources; Japanese and Korean scholarship terms are set by the applicable current call. [4][6][8]

Offer term	Questions to get answered
Study / employment status	Degree enrolment, non-degree research, employment contract, scholarship or another arrangement.
Duration	Start date, funded period, extension rules, probation or review milestones.
Coverage	Tuition/fees, stipend or salary, insurance, travel, bench or research costs, and any exclusions.
Conditions	Eligibility, progress reporting, work limits, renewal, leave, outside work and repayment terms.
Research terms	Supervisor, equipment, data access, IP ownership, publication review, collaboration and lab safety.
Living and residence	Current proof-of-funds/insurance needs, housing and the correct immigration route for you.
Career transition	What support exists for teaching, industry projects, internships, job search and work-status changes?

Before saying yes

Ask the admissions office, graduate school, supervisor or scholarship contact to point to the governing written terms. Save the exact call and your written answers. If a term affects intellectual property, employment, tax, work authorization or family plans, get qualified advice for your own circumstances.

> **A funded label is not a full budget**
> A scholarship or position may cover some costs and not others. List the actual amounts and dates from the offer and current official information before deciding. This guide contains no tuition, stipend, salary or living-cost figures.

05 / THREE-COUNTRY CAREER CHECKS

Plan the move from research to engineering work

Prepare for two outcomes while you research: continuing in academia and moving to industry or an applied research organization. A doctorate demonstrates research depth; it does not automatically establish fit for every engineering role. Translate your work into the problems a particular team is hiring to solve.

Country	Career / residence check	Useful preparation
Germany	Use your project, degree and methods to target employer tasks or research roles. DAAD lists doctoral research settings beyond universities, including research organizations and industry. If graduating in Germany, check the current residence process and apply in time. A regulated title or work route is a separate authority decision. [1, 5, 11]	Build one research proof and one operations, design, data or validation artifact. Match each to current vacancy language.
Japan	Map lab work to specific engineering functions and employer requirements. Before full-time work, the status of residence must allow it. Study in Japan directs students to the change-of-status procedure; confirm current eligibility and required documents with Immigration Services Agency. [9]	Check language used in actual target job descriptions, qualifications and contract. Do not assume a student status permits full-time work.
South Korea	Use current university and employer career channels. Check the official job-seeking status route and the relevant Hi Korea or Korea Visa Portal instructions before graduate status expires. Eligibility and conditions can differ by case. [10]	Document your technical evidence, target role and work-permission questions; confirm each with the employer and immigration authority.

Visa and employment rules are individual

Nationality, degree, host, job content, salary, timing and current law can all matter. These pages are navigation aids only. Ask the responsible immigration authority, university and employer before making a decision; never rely on this PDF alone.

06 / SHOW THE WORK, METHOD AND LIMIT

Translate research into an engineering story

For each research project, prepare a short technical explanation that a non-specialist engineering manager can follow and a specialist can probe.

Story element	Explain it in plain, testable language
Problem	What system or user need led to the question?
Contribution	Which analysis, design, experiment, model or instrument did you personally own?
Evidence	How did you validate the method? Which controls, baselines, datasets or test conditions matter?
Result	What did the work show? Include uncertainty, failed approaches or limitations.
Transfer	Which engineering task could use this method, and what would still need to be learned?

Illustrative example - structure only, not a real candidate outcome:

Illustrative research-to-industry summary

Investigated drift in a temperature-sensing setup; designed a calibration comparison, logged repeat runs and identified conditions where the model was less reliable. For a manufacturing role, I would first confirm sensor interfaces, required tolerances and the production validation process.

Keep the next career step concrete

- Choose a role family and collect live job descriptions from employers you could actually join.
- Map their tasks to your own work and name evidence for each match.
- Fill a gap with a permitted project, course, language step or industry collaboration.
- Ask the university career office what services are available to your exact student or graduate status.
- Record real application feedback, then revise one specific part of your materials.

The example above is fictional and contains no performance result. Never reuse it as your own experience.

07 / PROGRESS WITHOUT FALSE TARGETS

A review cycle for applications

Cycle-sensitive study and scholarship applications need a different rhythm from job search. Choose dates from the actual call, then review the work in short, evidence-based steps.

Sprint	Action
Screen	Confirm the program, degree level, supervisor/route, country, language and exact next deadline.
Assemble	Collect official documents, request references, prepare a tailored proposal and verify translations.
Submit	Use the current portal or embassy/university route; save confirmation and a copy of every file.
Prepare	Plan for an interview or examination only if the call requires it; prepare to explain your own research.
Review	Record the actual decision, feedback and next permitted step. Do not infer a general admissions rule from one result.

Open questions before committing

- What exactly is the degree, research status or job arrangement?
- Who supervises the work, and how are progress, authorship and conflict handled?
- What does the award or contract pay for, and what remains my responsibility?
- What are the live student, researcher and post-graduation residence rules?
- Which job families connect to my research, and what proof or language do their employers ask for?
- What is the official next action, and what is my backup if admission or funding does not happen?

Stop / change / continue

Continue when the official route, funding and research fit are clear. Pause when a critical term is missing. Change the shortlist when the program's work does not match your question or career evidence. This is a decision tool, not a prediction of acceptance.

PRIMARY SOURCES

Official source register / 1 of 2

Direct titles and links are included so you can recheck the rule or application page. Source pages can change; check the issuing body again immediately before acting.

[1] DAAD - Ways to Your PhD
URL: https://www.daad.de/en/studying-in-germany/phd-studies-research/ways-to-your-phd/
Use: Germany's individual and structured doctoral routes, and settings for doctoral research. | Accessed 11 Oct 2026.

[2] DAAD - Doing a PhD in Germany
URL: https://www.daad.de/en/studying-in-germany/phd-studies-research/phd/
Use: Starting point for Germany doctoral study, requirements and career planning. | Accessed 11 Oct 2026.

[3] Research in Germany - Find a PhD position in Germany
URL: https://www.research-in-germany.org/en/your-goal/phd/find-your-phd-position.html
Use: Official search-route overview, including the PhDGermany database and application information. | Accessed 11 Oct 2026.

[4] DAAD - How to finance your PhD
URL: https://www.daad.de/en/studying-in-germany/phd-studies-research/finance-your-phd/
Use: Funding routes and official scholarship/funding search starting points. | Accessed 11 Oct 2026.

[5] Study in Japan - Graduate Schools
URL: https://www.studyinjapan.go.jp/en/planning/learn-about-schools/graduate-schools/
Use: Graduate research-student distinction, common application documents and school-specific selection. | Accessed 11 Oct 2026.

[6] Study in Japan - Japanese Government (MEXT) Scholarship
URL: https://www.studyinjapan.go.jp/en/planning/scholarships/mext-scholarships/
Use: Research-student routes and embassy or university recommendation; follow latest guidelines. | Accessed 11 Oct 2026.

PRIMARY SOURCES

Official source register / 2 of 2

Direct titles and links are included so you can recheck the rule or application page. Source pages can change; check the issuing body again immediately before acting.

[7] Study in Korea - Korea's Education System
URL: https://www.studyinkorea.go.kr/en/plan/educationSystem.do
Use: Graduate degree structure and the need to check each school's admission criteria. | Accessed 11 Oct 2026.

[8] Study in Korea - 2026 GKS Graduate Degree Notice
URL: https://www.studyinkorea.go.kr/en/plan/gksNoticeRead.do?bbsId=BBSMSTR_000000000461&nttId=4420
Use: Published 2026 notice illustrates embassy and university tracks; it is not a claim that the cycle is open. | Accessed 11 Oct 2026.

[9] Study in Japan - Changing Status of Residence
URL: https://www.studyinjapan.go.jp/en/work-in-japan/immigration-procedures/
Use: Change of residence status before full-time work and checking job-seeking conditions. | Accessed 11 Oct 2026.

[10] Study in Korea - Residence Eligibility and Stay Duration (D-10)
URL: https://studyinkorea.go.kr/ko/life/residenceAndStayInfo.do?tab=job-seeker-visa
Use: Official job-seeking status overview; confirm live details with Hi Korea or Korea Visa Portal. | Accessed 11 Oct 2026.

[11] Make it in Germany - Prospects after graduation
URL: https://www.make-it-in-germany.com/en/study-vocational-training/studies-in-germany/prospects-after
Use: Germany graduate routes and the need to apply for the appropriate residence title. | Accessed 11 Oct 2026.

Correction and accessibility route: contact.html. If a rule or link appears out of date, report the source title and section so the guide can be reviewed.

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